SketchFlow Operation Manual

Oct 29, 2026 ·

1. Introduction

SketchFlow is a Windows desktop application that combines PDF markup with fire sprinkler and hydrant hydraulic calculations, so designers can sketch pipework over drawings, solve the network and issue a report in one tool.

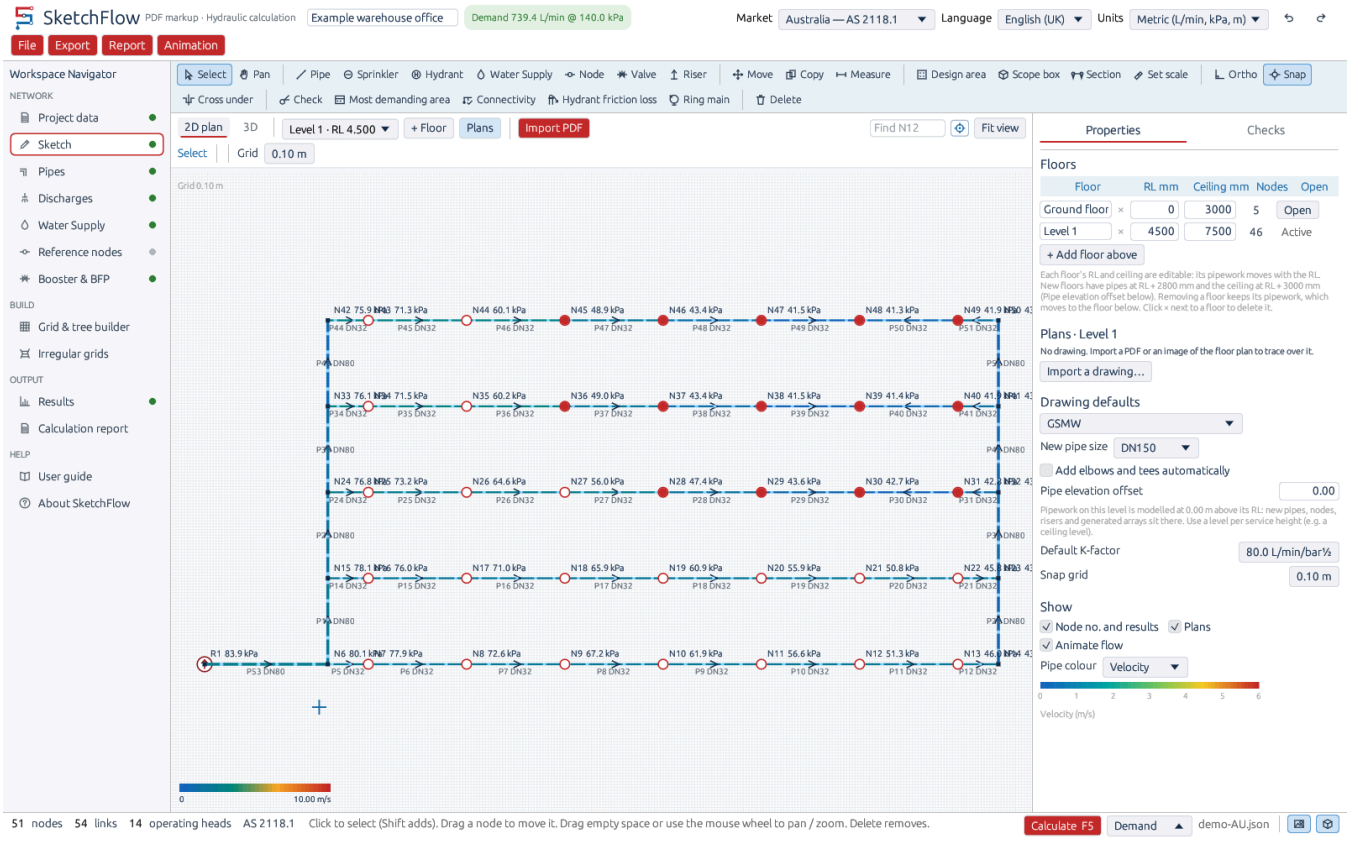


Figure 1 — SketchFlow main window: a calculated sprinkler array with node pressures, flow arrows and pipe sizes on the 2D plan.

Purpose of this manual. It explains how to install SketchFlow, how each part of the application works, and how to request and apply a licence activation key.

Intended users. Fire protection engineers, hydraulic designers, building services designers and reviewers working on sprinkler and hydrant systems.

1.1 Product at a glance

Item	Detail
Product	SketchFlow (listed on the Store as "Sketch Flow")
Version	1.0.23
Platform	Windows desktop (64-bit)
Distribution	Microsoft Store — Sketch Flow (Store ID 9PF4CZDJ17RV)
Price	Free download; runs in Demo mode until a licence key is activated
Design standards (Market)	Australia — AS 2118.1 · New Zealand — NZS 4541 · United States — NFPA 13 · Canada — NFPA 13 · United Kingdom — EN 12845 · France — EN 12845
Units	Metric (L/min, kPa, m) · Metric (L/min, bar, m) · US (gpm, psi, ft)
Connectivity	Works offline; internet is used only for the optional update check
Languages	English and French interface
Calculation methods	Hazen-Williams or Darcy-Weisbach friction loss, global-gradient network solver
Developer	Mohamed Mustafa

1.2 What SketchFlow does

- Imports PDF drawings and images as underlays and lets you sketch pipework directly over them in 2D.
- Builds multi-floor networks with risers and droppers, viewable in an interactive 3D view.
- Generates sprinkler grids and trees automatically (end-fed, centre-fed, grid and irregular layouts).
- Solves looped and branched networks, then checks the result against the available water supply.

- Identifies the governing outlet, the most demanding area and hydrant friction losses.
- Produces text and PDF hydraulic reports and exchanges data with common industry file formats.

2. System Requirements and Installation

SketchFlow installs from the Microsoft Store in a few minutes and needs no internet connection after installation.

2.1 Requirements

Requirement	Specification
Operating system	Windows 10 or Windows 11, 64-bit
Account	Microsoft account to download from the Store
Internet	Needed only to download the app and to email an activation request
Email client	A default mail app makes "Send Request" one click; otherwise use "Copy Info"
Storage location for licence	`%APPDATA%\SketchFlow\` (created automatically)

2.2 Installing from the Microsoft Store

1. Open the SketchFlow page on the Microsoft Store, or search "Sketch Flow" in the Microsoft Store app.
2. Click **Get** (or **Install**) and sign in with your Microsoft account if asked.
3. When installation finishes, launch SketchFlow from the Start menu.
4. The application opens in **Demo mode**. You can explore every screen immediately; see Section 8 to activate a full licence.

2.3 Updating and uninstalling

- Updates are delivered automatically through the Microsoft Store.
- Reinstalling SketchFlow on the same computer keeps the existing licence activation.

- To move a licence to a different computer, use **Deactivate on this PC** first, then request a new key for the new machine (Section 8).

3. User Interface Overview

The SketchFlow window is organised around a central drawing area, with menus at the top, project navigation on the side and live status at the bottom.

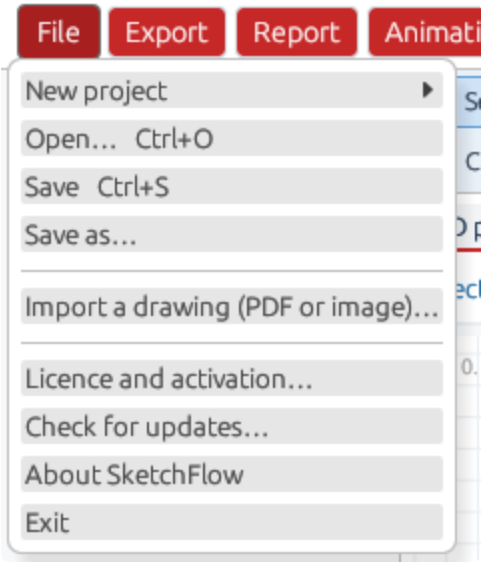


Figure 2 — The File menu, including Licence and activation and Check for updates.

Area	Location	What it does
Top bar	Top of window	Project name, calculation status (e.g. *Demand 739.4 L/min @ 140.0 kPa*), licence badge, **Market** (design standard), **Language**, **Units**, undo and redo
Menus	Under the logo	**File** (new, open, save, import drawing, licence and activation, check for updates, about), **Export**, **Report**, **Animation** (export flow animation as GIF)
Workspace Navigator	Left panel	Network: Project data, Sketch, Pipes, Discharges, Water Supply, Reference nodes, Booster & BFP. Build: Grid & tree builder, Irregular grids. Output: Results, Calculation report. Help: User guide, About

Area	Location	What it does
Toolbar	Above the plan	Select, Pan, Pipe, Sprinkler, Hydrant, Water Supply, Node, Valve, Riser, Move, Copy, Measure, Design area, Scope box, Section, Set scale, Ortho, Snap, Cross under, Check, Most demanding area, Connectivity, Hydrant friction loss, Ring main, Delete
Plan bar	Above the plan	2D plan / 3D switch, floor selector, + Floor, Plans, Import PDF, Find, Fit view
2D plan / 3D view	Centre	Sketching area over drawings; 3D model with view cube
Properties / Checks	Right panel	Floors, plans, drawing defaults, display options and pipe colour legend; design checks tab
Status bar	Bottom	Node, link and operating-head counts, standard, hints, **Calculate (F5)** and calculation mode

Interface language. SketchFlow's screens are available in English and French.

Keyboard shortcuts. A full list is available from the Shortcuts dialog in the Help menu.

4. Setting Up a Project

A SketchFlow project is prepared before sketching by entering project data, floors, discharge defaults, materials and the water supply.

4.1 Recommended workflow

1. **Create or open a project** — File → New or File → Open. Projects are saved as JSON files.
2. **Enter project data** — complete the Project data tabs: project details, system type and design basis.
3. **Define floors** — add levels and their reduced levels (RL).
4. **Set discharge defaults** — enter the default sprinkler and hydrant flows and pressures.
5. **Confirm materials** — check pipe materials, internal diameters and roughness values.
6. **Enter the water supply** — input the supply curve (pressure vs flow) at the connection point.

7. **Sketch the network** (Section 5), then **calculate** (Section 6).

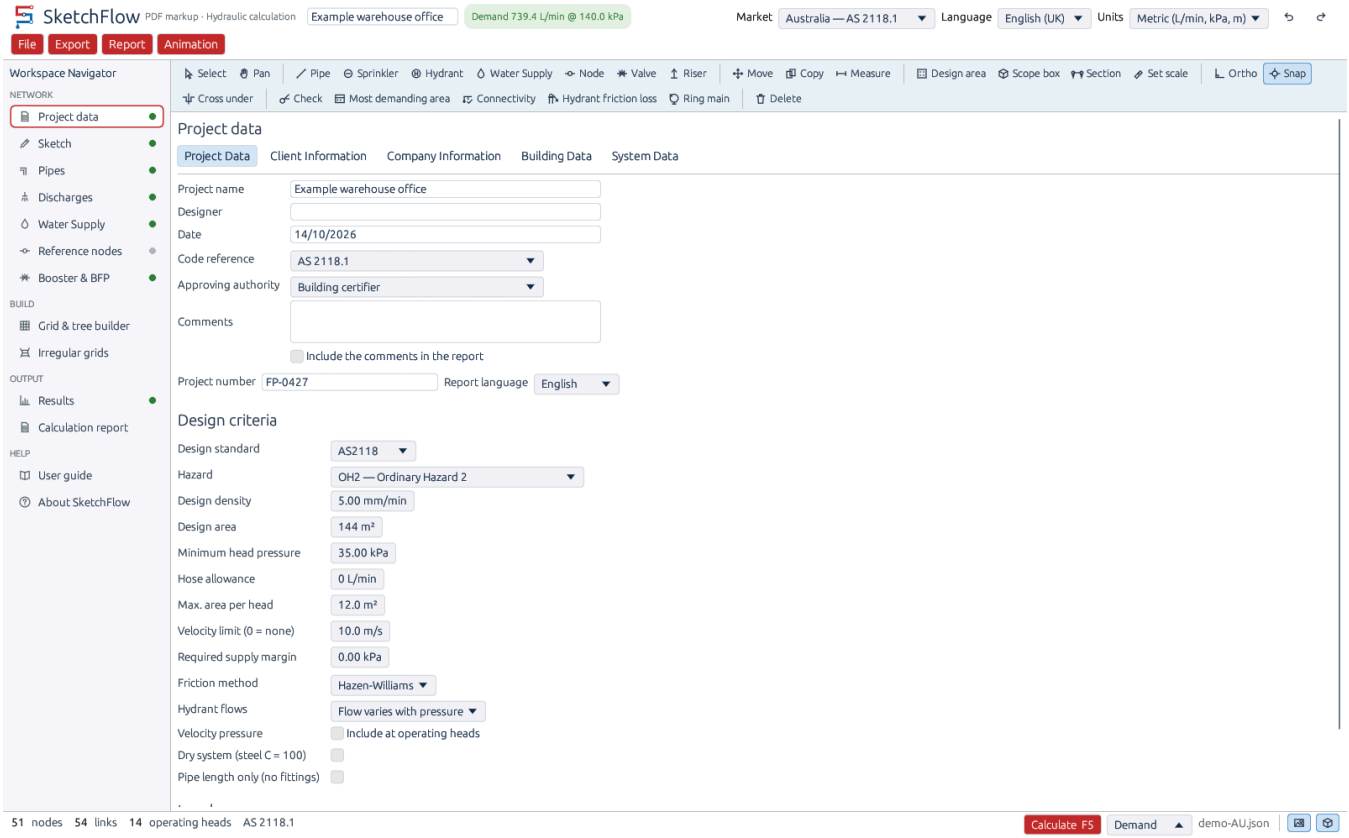


Figure 3 — Project data: project details, code reference, approving authority and design criteria (hazard, density, design area, minimum head pressure, friction method).

4.2 Floors

- A new level is created 3 m above the current top level; adjust as required.
- Each floor has an offset that is added to node elevations on that floor.
- Changing a floor's RL moves all nodes on that floor with it.
- Removing a floor keeps its pipework, which moves to the floor below.

4.3 Discharges

- The **Defaults** row sets the standard design flow for sprinklers and hydrants. Changing it applies to every item in that table.
- New or converted outlets take the defaults automatically.
- Hydrants without a specified design flow are filled from the defaults when you calculate.

- **Hose streams (USA and Canada):** inside and outside hose stream allowances are entered on the Discharges page and added at the supply only.

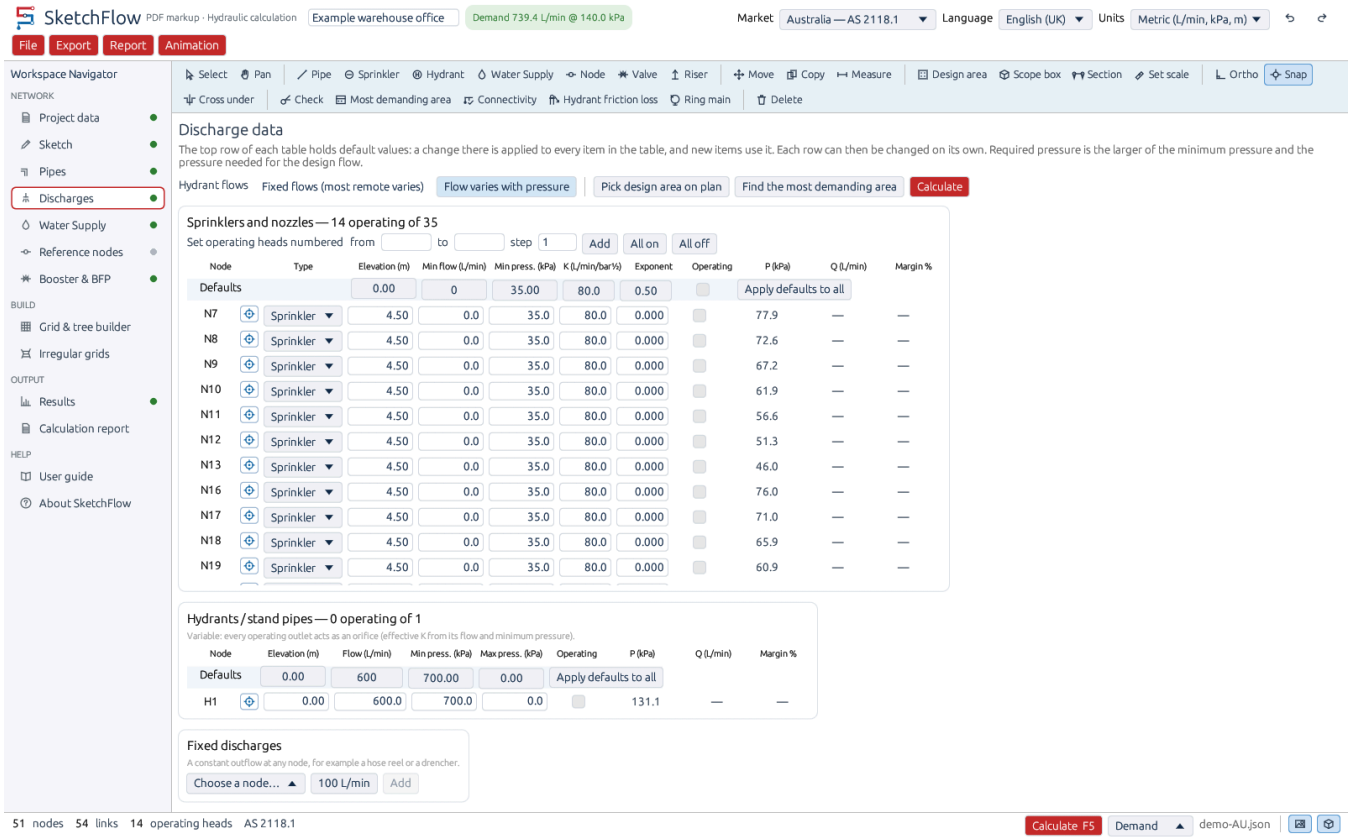


Figure 4 — Discharge data: the Defaults row at the top of each table, operating sprinklers with K-factor and minimum pressure, and hydrants/standpipes below.

4.4 Materials and pipework library

- The Materials page holds pipe types, nominal and internal diameters and friction coefficients.
- The pipework library can be exported to and imported from Excel to share company standards across projects (export requires a licence).
- Choose Hazen-Williams or Darcy-Weisbach friction loss to suit the project standard.

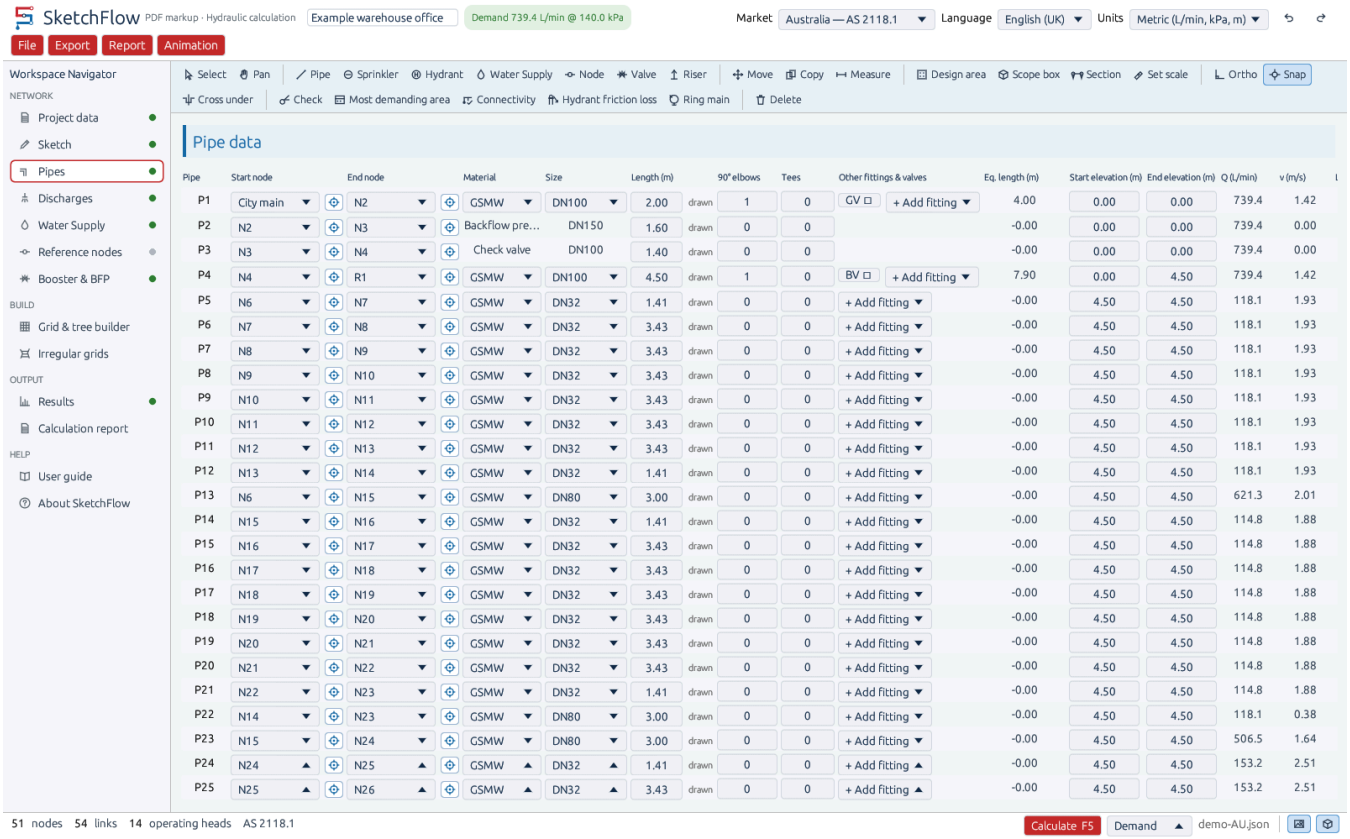


Figure 5 — Pipe data: start and end nodes, material, size, length, elbows, tees, fittings and valves, equivalent length, elevations, flow and velocity for every pipe.

4.5 Water supply

- Enter the supply curve from the flow test or authority data.
- SketchFlow adds the hose allowance to the outlet flow to obtain the total demand flow.
- Available pressure is read from the curve at that total flow; the margin equals available minus required pressure.



Figure 6 — Water supply: supply curve test points and the supply-and-demand graph, showing demand 739.4 L/min @ 140.0 kPa against 538.8 kPa available (margin 398.7 kPa).

5. Sketching the Pipework

Pipework is drawn directly on the 2D plan, over imported drawings, with snapping and tracking that keep the network connected as you draw.

5.1 Importing drawings

- Use **Import PDF** on the toolbar (or File menu) to load architectural or services drawings as underlays. Images are also supported.
- Calibrate the scale with a known dimension, then add dimensions and reference nodes as needed.

5.2 Drawing tools

Tool	How to use it
Pipe / main	Click to place points; double-click the last point to finish the run
Node and outlet	Place sprinklers, hydrants and junctions; set K-factor and design flow in Properties
Tee	Click on an existing pipe to split it and create a tee
Riser / dropper	Use the Riser tool, or right-click a node or pipe; shown on plan as a circle with an up or down arrow
Right-click menu	Context actions on nodes and pipes, including riser creation and conversion
Find	Locate a node or pipe by its identifier and jump to it
Dimensions and reference nodes	Annotate the plan and mark fixed points

5.3 Snap and tracking

- The pointer snaps to nodes, pipe midpoints and points on pipes.
- With **Snap** on, a point aligned with another node takes that node's X or Y coordinate (tracking), including the intersection of two alignments.
- Each new connection flashes green with the message **Connected**, confirming the pipework is joined.

5.4 Automatic grid and tree builder

The **Grid and tree builder** generates sprinkler layouts in seconds:

- **Layouts:** end-fed, centre-fed, grid and irregular.
- **Fit to room:** pick a room on the plan and SketchFlow places the grid inside it.
- **Irregular grids:** a dedicated window and plan overlay handle non-rectangular mains.

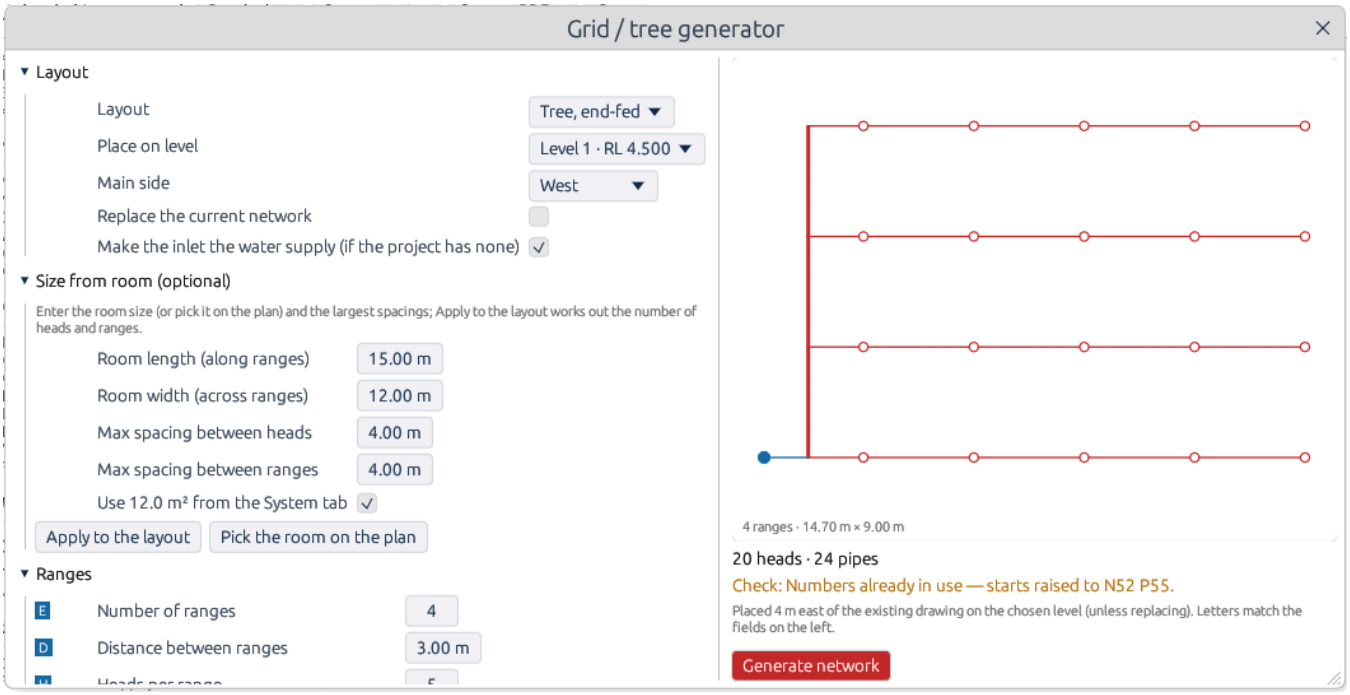


Figure 7 — Grid / tree generator: choose the layout, level and main side, size it from the room, preview the result and click Generate network.

5.5 3D view

- Switch to the 3D view to check risers, droppers and floor-to-floor connections.
- Use the view cube to jump to top, front, side and isometric views.

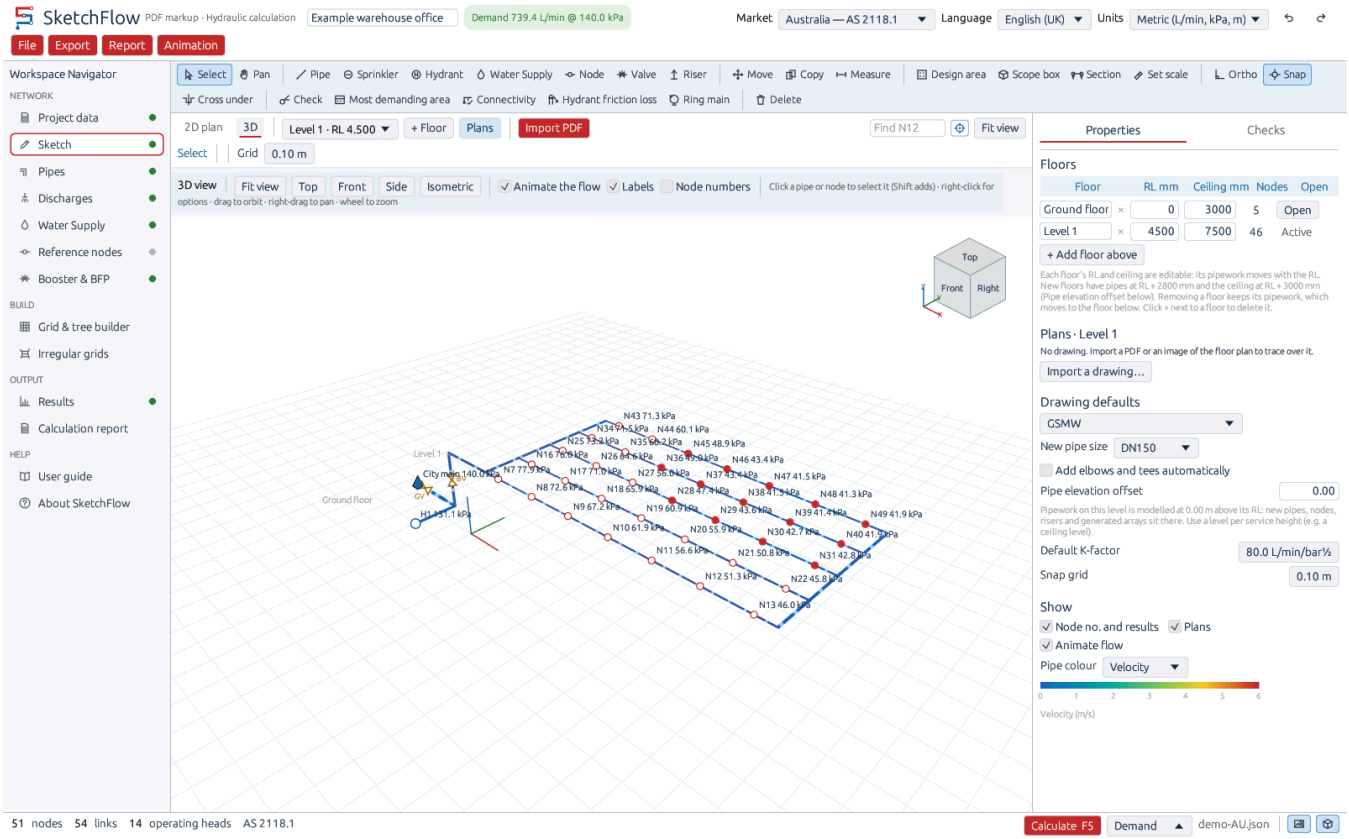


Figure 8 — 3D view: both floors, the riser and the Level 1 array with node pressures; Fit view, Top, Front, Side and Isometric buttons and the view cube.

6. Hydraulic Calculation and Results

One click on **Calculate** solves the whole network — looped or branched, single or multi-floor — and opens the Results view with the supply check and design checks.

6.1 Running a calculation

1. Confirm every outlet has a design flow (hydrants without one are filled from the Discharges defaults).
2. Click **Calculate**.
3. SketchFlow solves flows and pressures using the global-gradient method with Hazen-Williams or Darcy-Weisbach friction loss.
4. Hose stream allowances (where entered) are added at the supply.
5. Results open automatically; any blocking issue is shown as a clear message.

6.2 Results view

Output	What it shows
Results cards	Total flow, required pressure, governing outlet and key values at a glance
Supply check banner	Pass or fail of the system demand against the water supply, with the pressure margin
Supply graph	Supply curve plotted against the system demand point
Water supply calculation table	Outlet flow + hose allowance = total flow; available pressure at that flow; margin
Design checks	Governing (most unfavourable) outlet and compliance checks
Most demanding area	Identifies the hydraulically most demanding design area
Flow view	Flow arrows and flow labels on every pipe, with optional flow animation
Pipes, Reference nodes tables	Full node and pipe results, each row with a locate button that jumps to the item on the plan



Figure 9 — Results: summary cards (supplied flow, sprinklers operating, governing flow, maximum velocity), the green water supply check banner and the flow view with per-pipe flows.

6.3 Reading the supply check

- **Total flow** = outlet flow + hose allowance.
- **Available pressure** is read from the supply curve at the total flow.
- **Margin** = available pressure − required pressure. A positive margin means the supply is adequate.

7. Network Analysis, Reports and Exports

Beyond the main calculation, SketchFlow includes three network analysis tools and produces submission-ready reports.

7.1 Network analysis tools

Tool	Purpose
Connectivity	Checks that every outlet is connected to the supply and highlights disconnected pipework on the plan
Hydrant friction loss	Calculates friction loss from the supply to the most unfavourable hydrant and highlights the path
Ring main check	Tests a hydrant ring main with both legs open, then each leg alone with the other blocked; uses the selected pipe's loop or the loop with the most hydrants

If a loop is not connected to the supply, SketchFlow shows a clear message with a shortcut to the Connectivity tool.

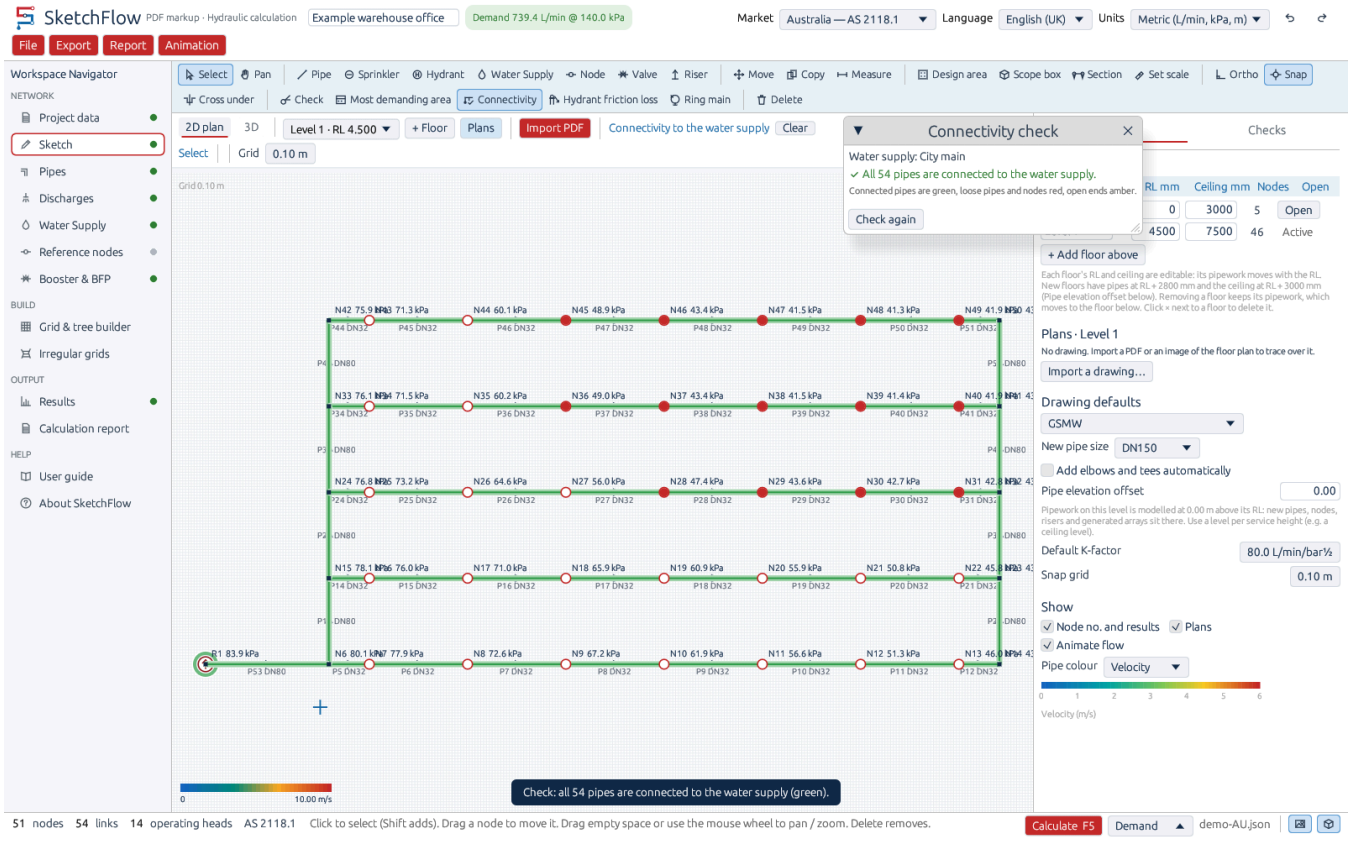


Figure 10 — Connectivity check: all 54 pipes connected to the water supply are shown green; loose pipes would show red and open ends amber.

7.2 Reports

- **PDF report** — a formatted hydraulic calculation report for submission and review, from the **Report** menu.
- **Text report** — a plain-text summary of inputs and results.
- **Command-line tool** — `sketchflow-cli` can run calc, report and pdf commands for batch work.

The Calculation report page has **Recalculate**, **Save as text…**, **Save as PDF…** and **Copy** buttons. The **Animation** menu can also save the Results flow view as a looping animated GIF for presentations.



Figure 11 — Calculation report: project details, design criteria, water supply, results and a design-checks table with Pass/Fail for each check.

7.3 File exchange and exports

Format	Use
Project file (.json)	Native SketchFlow project

Format	Use
.FIW, .DAT, .VDB	Exchange files compatible with common hydraulic design workflows
Excel (.xlsx)	Pipework library export and import (licensed feature)
Help as web page	Save the built-in user guide for offline reference

8. Demo Mode and Licence Activation

SketchFlow downloads free and runs in Demo mode; a licence key, issued for your email address and computer, unlocks the full application with no limits or watermarks.

8.1 Demo mode vs licensed

Feature	Demo mode	Licensed
Network size	Up to 10 nodes	Unlimited
Open larger projects	Yes, view only	Yes
Calculate	Up to 10 nodes	Unlimited
PDF and text reports	Yes, watermarked "DEMO - NOT LICENSED"	Yes, no watermark
Excel pipework library export	Locked	Available
All screens, tools and 3D view	Available to explore	Available

Licence editions: Trial (time-limited), Standard and Professional. Licences can be perpetual or issued for a fixed term.

8.2 How to request activation

1. Open SketchFlow and go to **File → Licence and activation…**
2. Type the email address the licence will be registered to. Check every character — the key only works with this exact email.

3. Note the **Machine ID** shown in the window (8 groups of 4 characters). It is generated automatically for this computer.
4. Click **Send Request**. Your email app opens a pre-filled message to **sales@ai-es.com.au** containing your email and Machine ID. Send it.
    - No email app set up? Click **Copy Info** and paste the details into an email from any mail service.
5. Include your company name, the edition required and your purchase or order reference, if you have one.

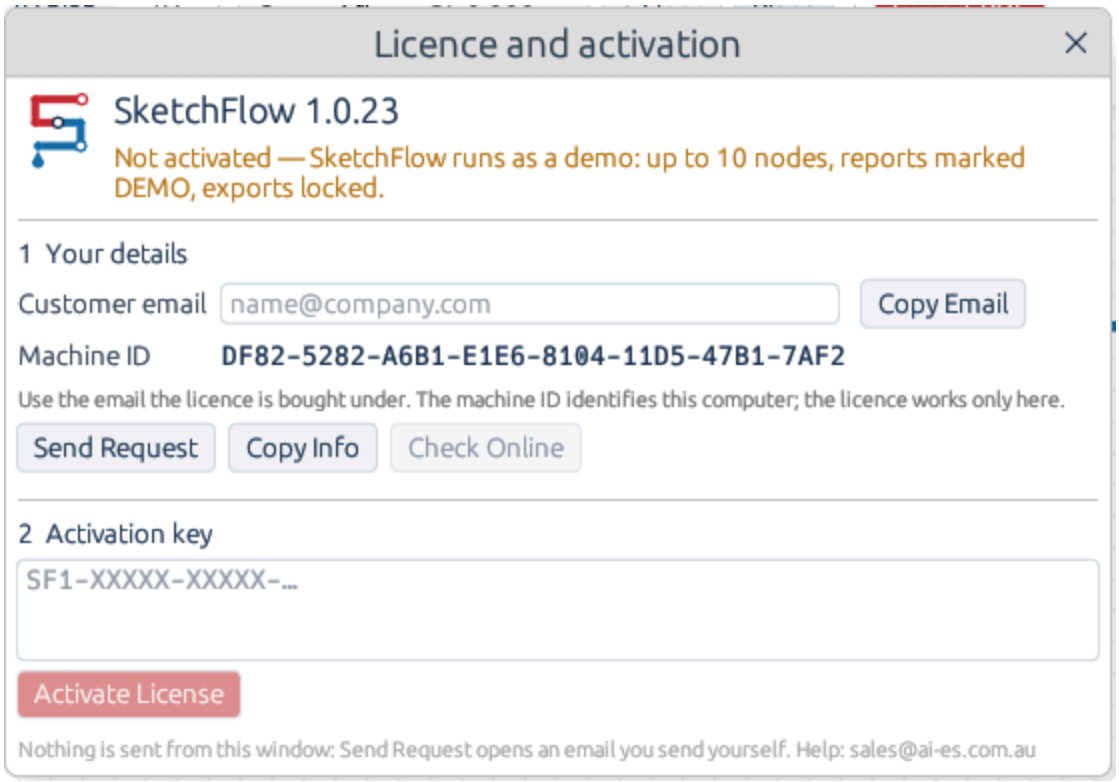


Figure 12 — Licence and activation window: step 1 (customer email, Machine ID, Send Request / Copy Info) and step 2 (paste the SF1- key and click Activate License). Nothing is sent from this window; Send Request opens an email you send yourself.

8.3 Activating your key

1. You will receive a licence key by email, beginning with `SF1-`.
2. Return to **File → Licence and activation…**
3. Paste the full key and click **Activate License**.
4. The status changes to licensed; all demo limits and watermarks are removed immediately.

8.4 Licence rules

- A key is bound to **one email address on one computer**.
- Reinstalling SketchFlow on the same computer keeps the licence.
- A new computer or a different email address needs a new key.
- To transfer, click **Deactivate on this PC** on the old computer, then request a key for the new one.
- The licence is stored encrypted at `%APPDATA%\SketchFlow\license.dat`; a copied or edited licence file will not work.
- No internet connection is needed to activate — the key is verified on your computer.

9. Troubleshooting and Support

Most issues are resolved by checking the activation details or network connectivity; for anything else, contact support with your Machine ID.

9.1 Common messages

Message or symptom	Cause	Solution
"Key is for another email or computer"	Email typed differently from the request, or key issued for another Machine ID	Re-type the email exactly as in the request; confirm the Machine ID matches
"Expired"	Trial or fixed-term licence has ended	Request a renewal or a perpetual licence
"Clock turned back"	System date set more than 2 days earlier than last use	Correct the Windows date and time, then restart SketchFlow
Cannot add more than 10 nodes	Running in Demo mode	Activate a licence (Section 8)
Calculate blocked on an opened project	Project exceeds the demo limit	Activate a licence
Report shows "DEMO - NOT LICENSED"	Running in Demo mode	Activate, then re-export the report
Loop not connected to supply	Gap in the pipework	Run **Connectivity** and join the highlighted pipes

9.2 Contact

- **Licensing and activation:** sales@ai-es.com.au
- **Download:** SketchFlow on the Microsoft Store
- **In-app help:** Help menu → User guide (searchable)

When contacting support, include your email address, Machine ID, SketchFlow version and a screenshot of any error message.

9.3 Quick reference

1. Install from the Microsoft Store.
2. Set up project data, floors, discharges, materials and water supply.
3. Import the PDF drawing and sketch the pipework (or use the grid builder).
4. Calculate and review Results and the supply check.
5. Run Connectivity, Hydrant friction loss or Ring main checks as needed.

6. Export the PDF report.
7. To unlock full use: File → Licence and activation… → Send Request → paste key → Activate License.